

Introdução

Curso: CFD Básico

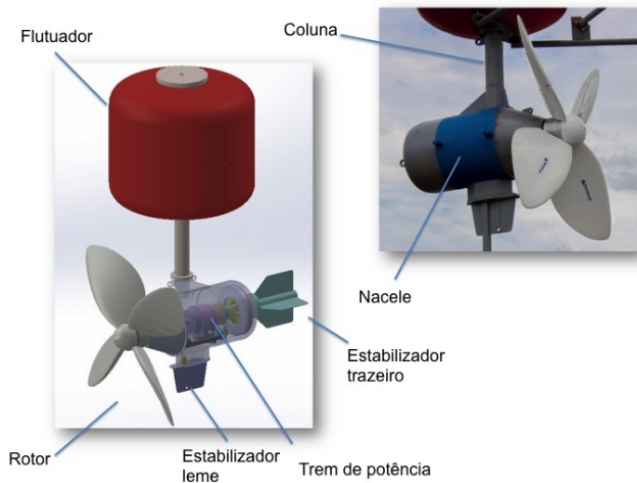
Professor: Adriano Possebon Rosa

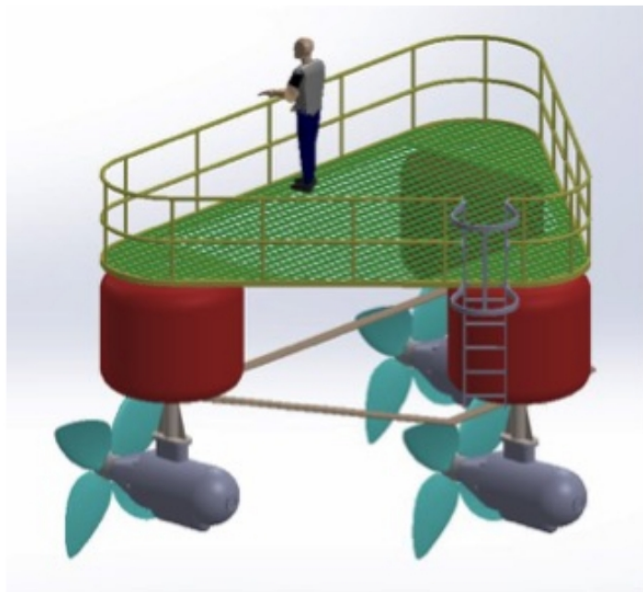
Departamento de Engenharia Mecânica
Universidade de Brasília

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- 2 Softwares CFD
- 3 OpenFOAM
- 4 Plano de Ensino
- 5 Material de Apoio
- 6 Trabalhos
- 7 Tarefa de Casa





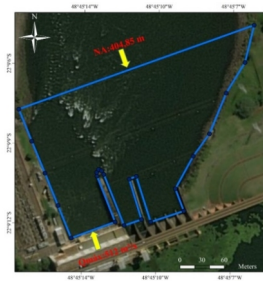
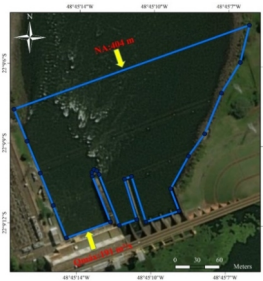


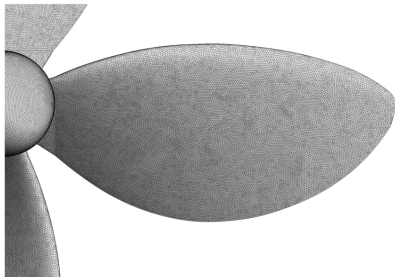
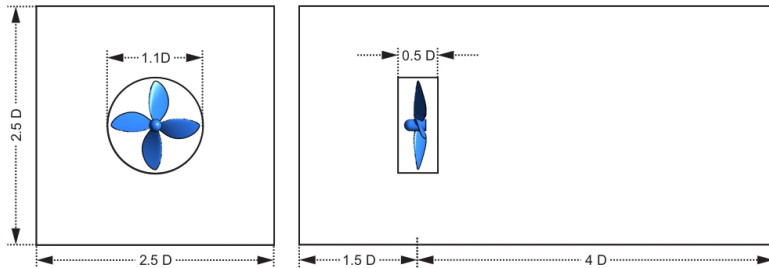


(a)



(b)





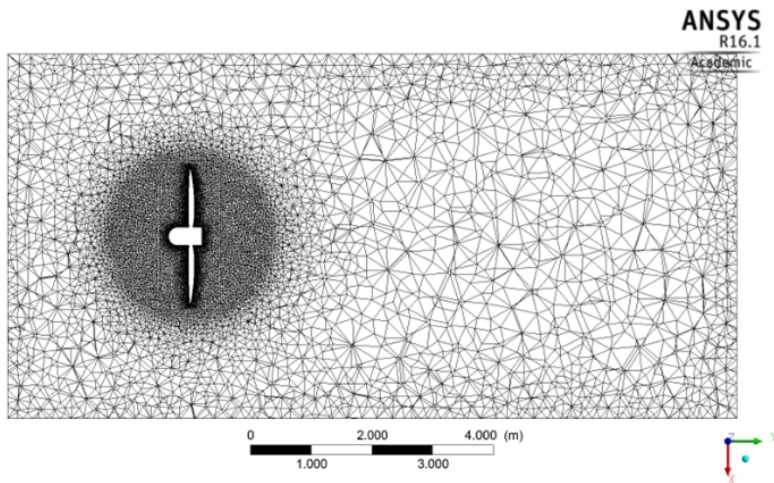


Figura 5.10: Malha 1 (426216 nós)

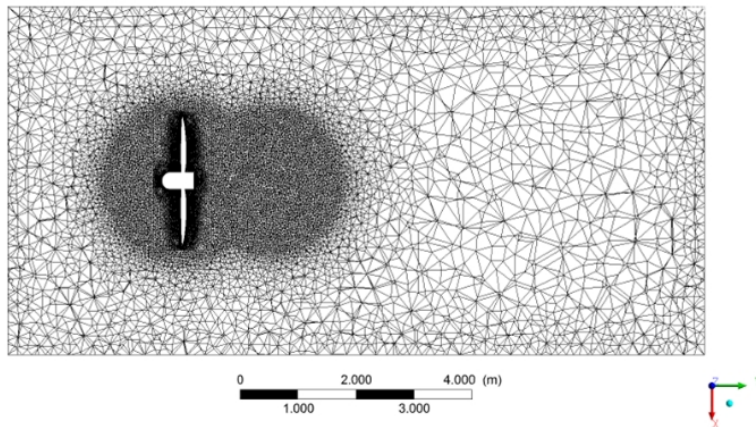


Figura 5.11: Malha 2 (1285254 nós)

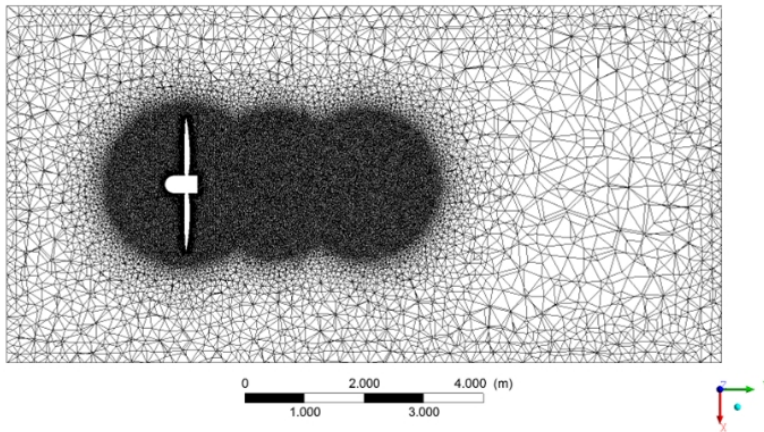
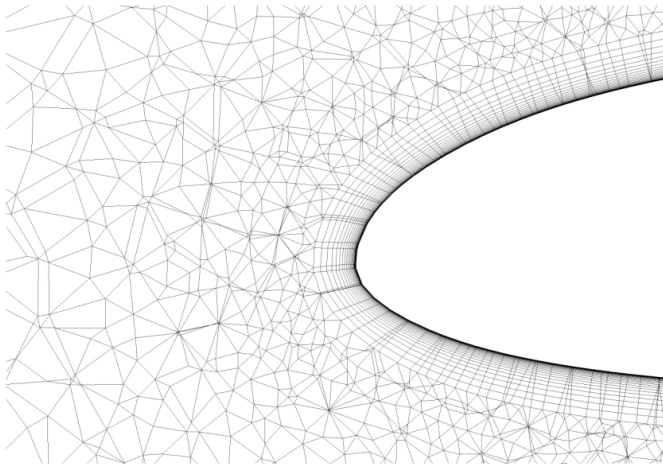
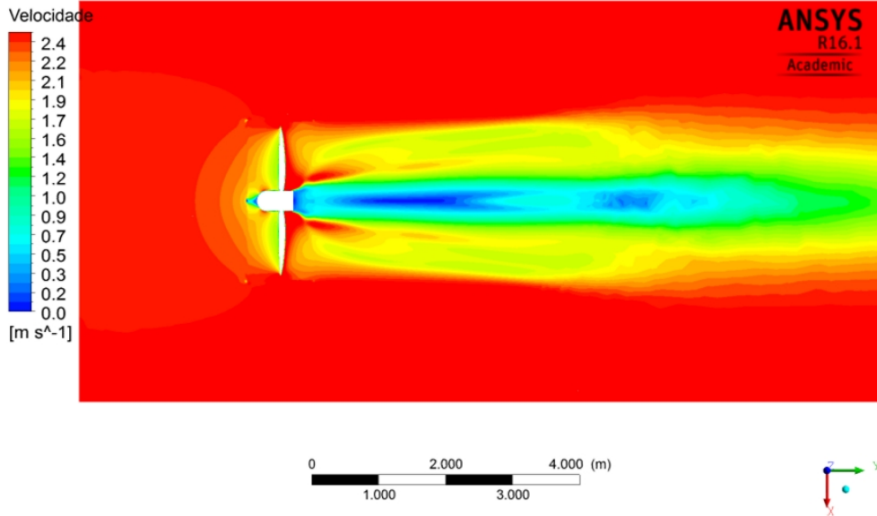


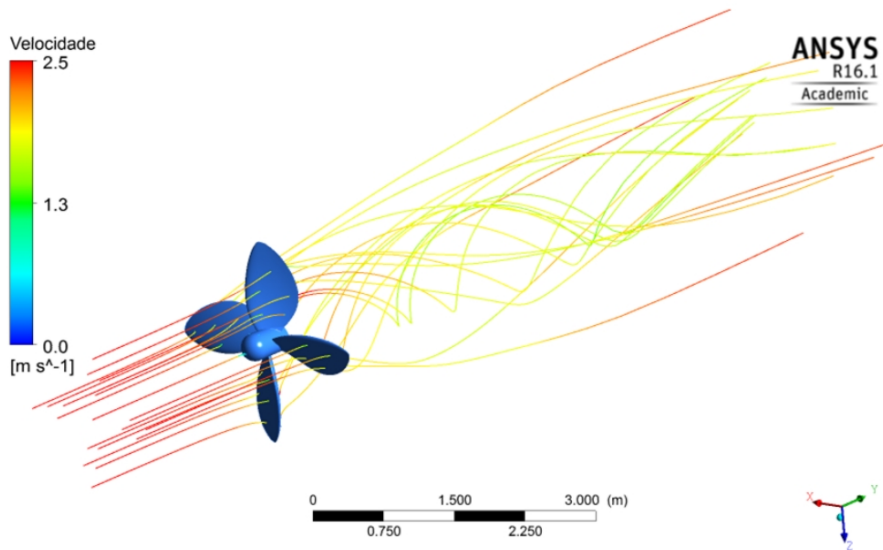
Figura 5.12: Malha 3 (5572162 nós)

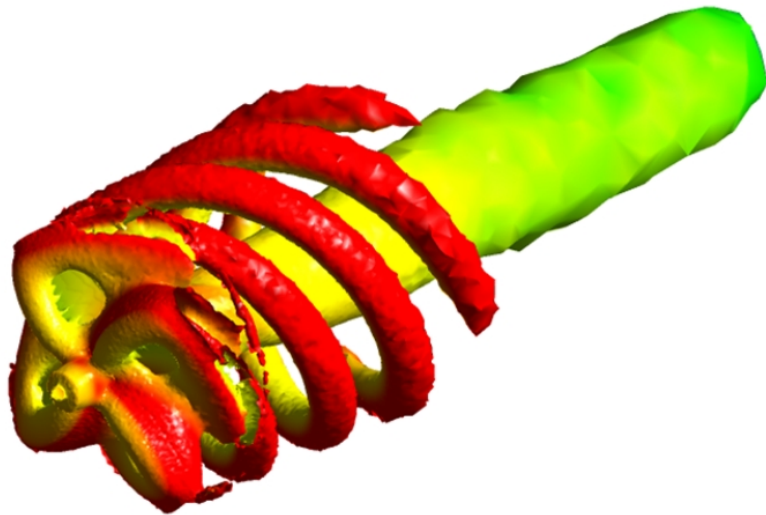
Tabela 5.3: Estudo de convergência de malha.

	Num. de nós	Refinamento esteira [m]	y+		Cp	$U_{y=10m}$
			Máximo	Médio		
Uma turbina						
Malha 1	0.426×10^6	2	323.62	4.681	0.381	1.5963
Malha 2	1.285×10^6	4	117.77	0.723	0.386	1.5499
<i>Malha 3</i>	5.572×10^6	6	1.57	0.383	0.38005	1.5496

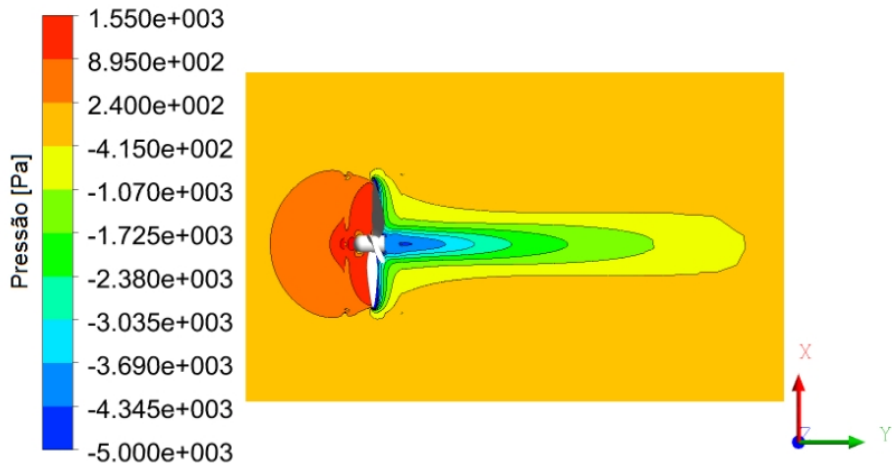








ANSYS
R16.1
Academic



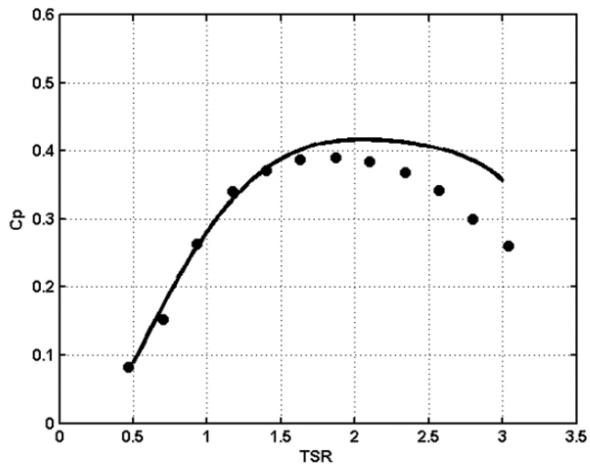


Figura 5.21: Curva de $C_p \times TSR$. — BEM ; • CFD

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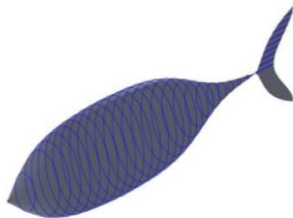


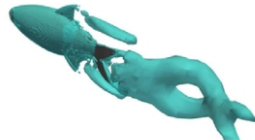
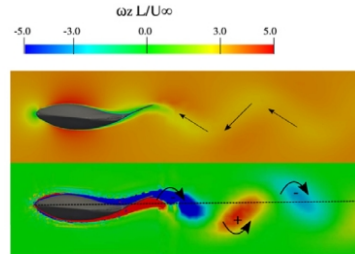
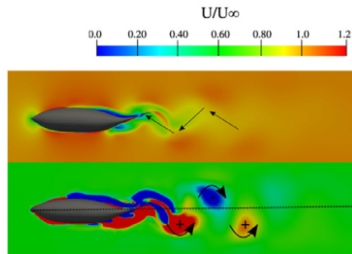
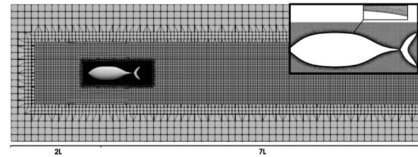
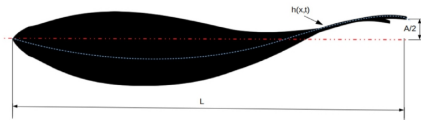
Three-dimensional viscous wake flow in fish swimming - A CFD study

Marianela M. Macias*, Igor F. Souza, Antonio C.P. Brasil Junior, Taygoara F. Oliveira

Campus Darcy Ribeiro, Asa Norte, 70910900. Energy and Environmental Laboratory, Mechanical Engineering Departament, Faculty of Technology. University of Brasilia, Brazil







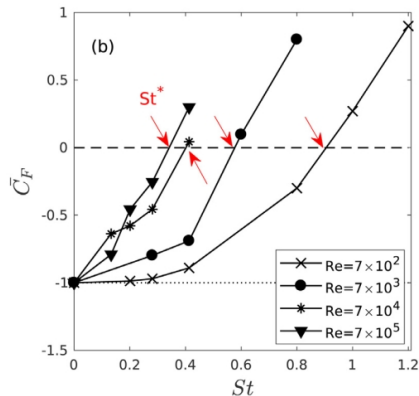
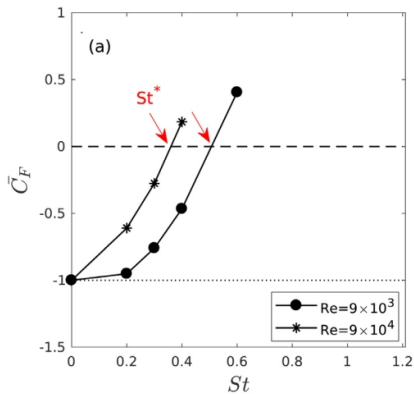


Fig. 5. $\bar{C}_F$ as a function of St for different Re . (a) Lambari; (b) Tuna fish. The red arrows mark the estimated equilibrium Strouhal. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

TECHNICAL PAPER



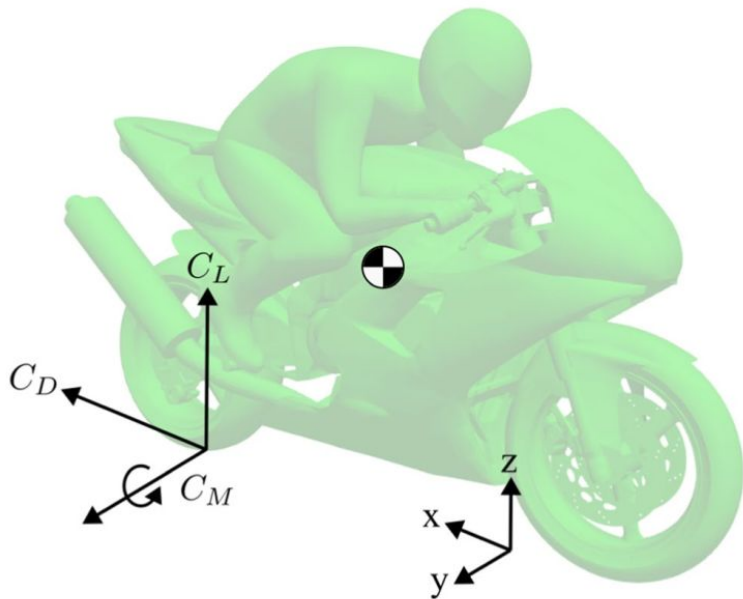
Numerical simulation of the aerodynamic effects of downforce-generating wings on trailing motorcycles: stability and safety considerations

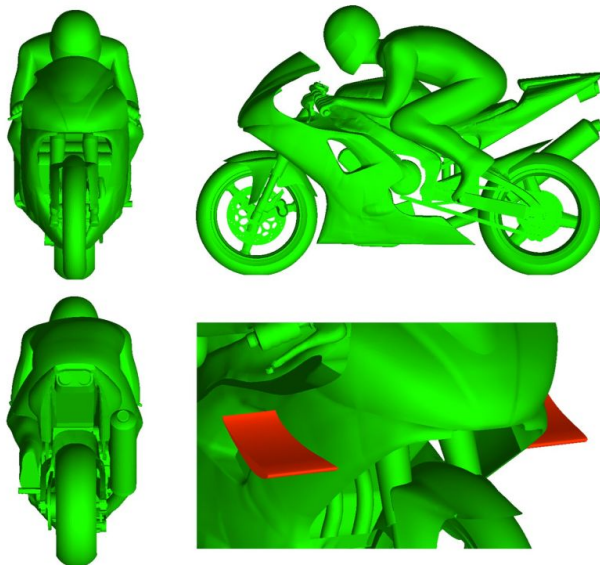
Braulio Gutierrez Pimenta¹  · Luís Paulo de Queiroz Moreira¹ · Adriano Possebon Rosa¹ · Roberto Francisco Bobenrieth Miserda¹ · Ana Luisa Pereira Maldonado²

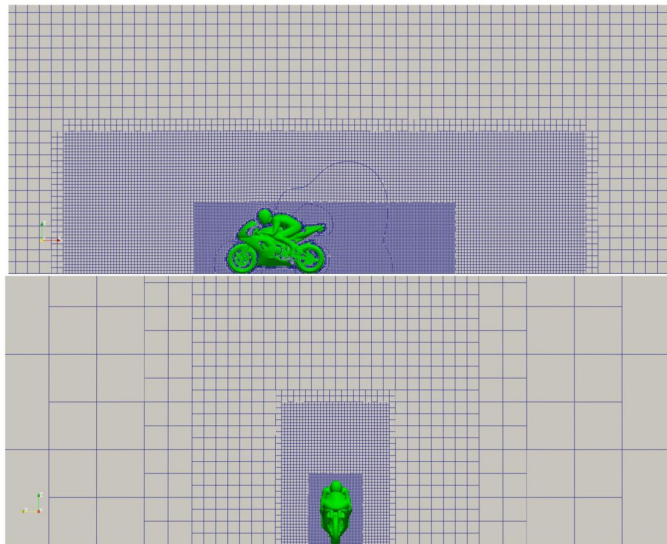
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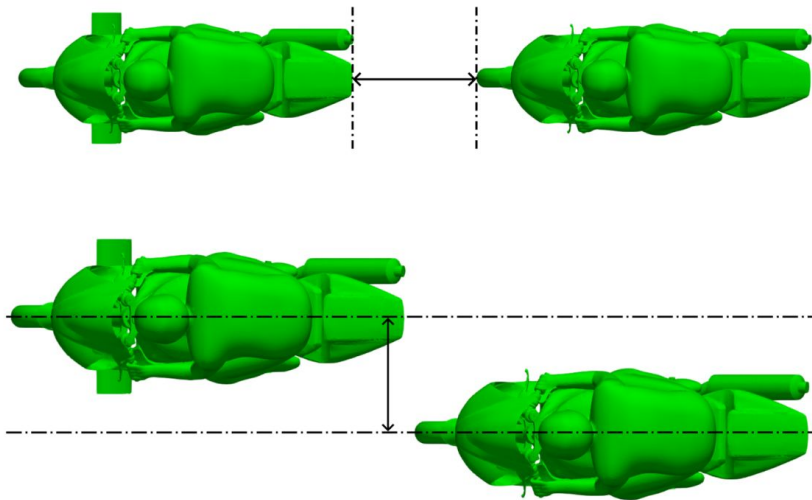
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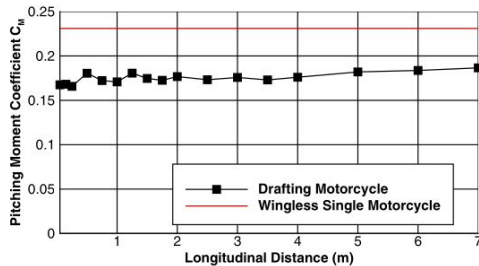
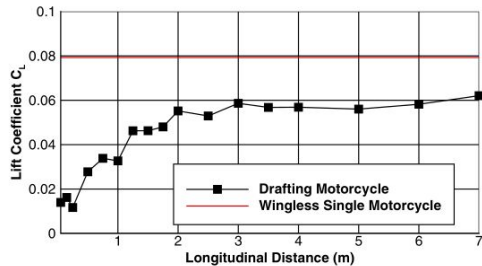
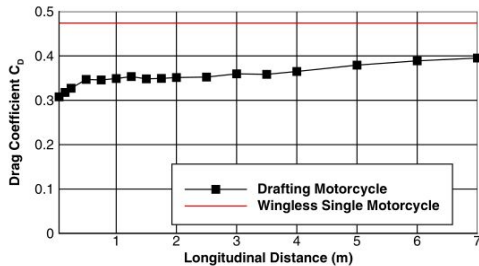


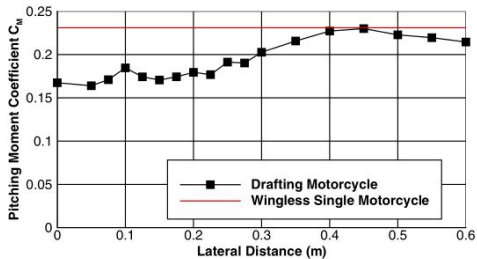
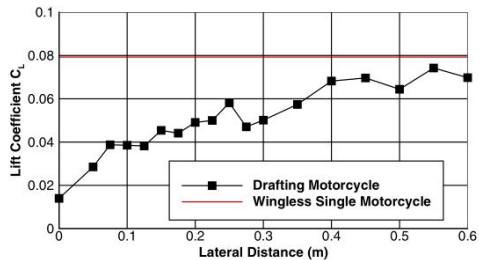
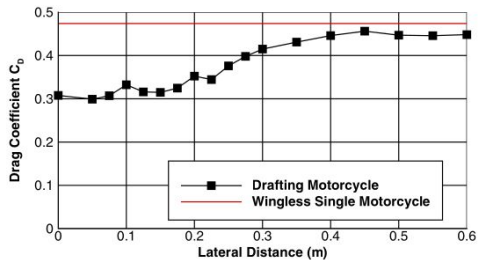


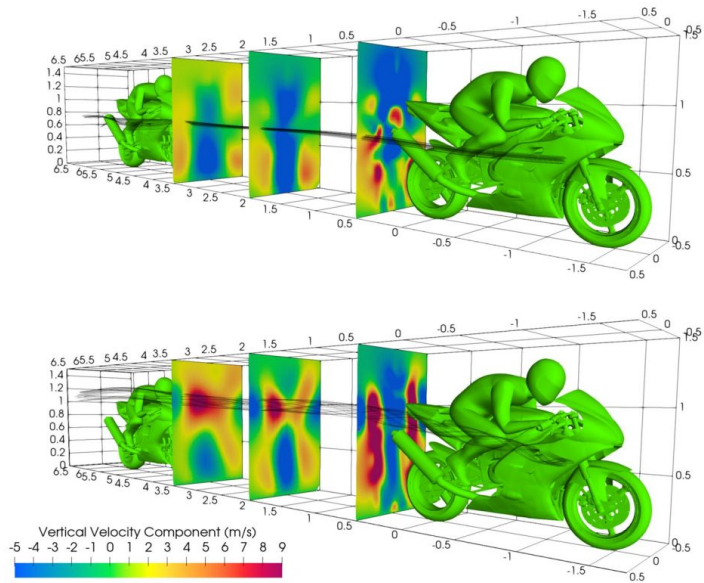








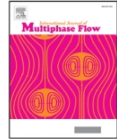






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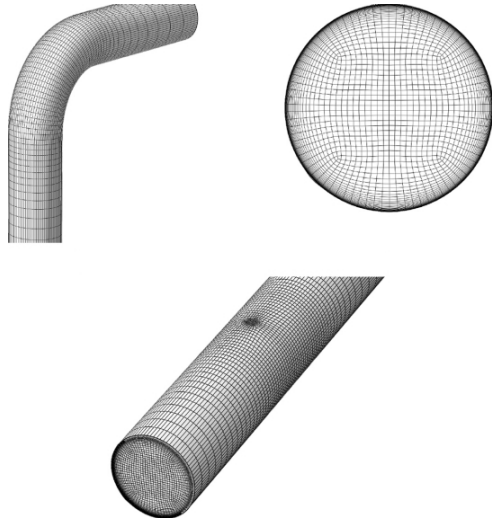
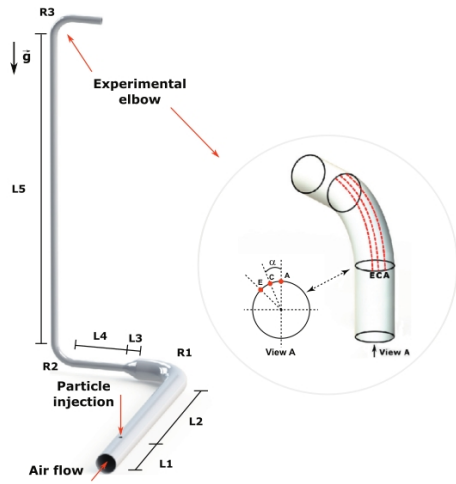
The role of inter-particle collisions on elbow erosion



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Vinicius Fagundes dos Santos^a

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^b School of Mechanical Engineering, Technological Federal University of Paraná, Av. dos Pioneiros, 3131 Bloco B, Londrina, Paraná 86036-370, Brazil



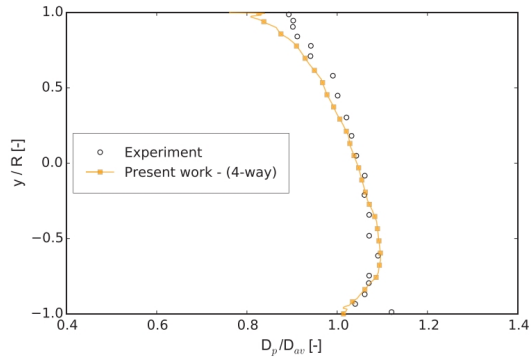
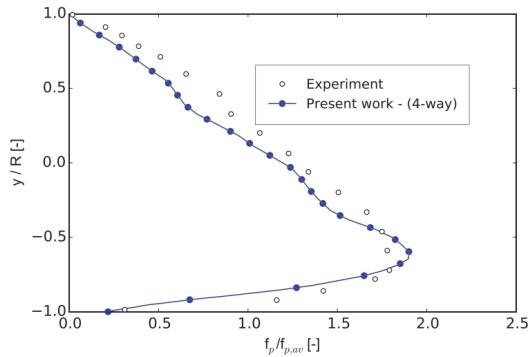
$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1)$$

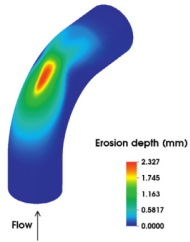
$$\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial (\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[(\mu + \mu_t) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] + S u_{ip} + \rho g_i \quad (2)$$

$$\frac{dx_{pi}}{dt} = u_{pi} \quad (3)$$

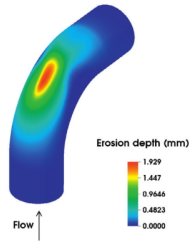
$$m_p \frac{du_{pi}}{dt} = m_p \frac{3\rho C_D}{4\rho_p d_p} (u_i - u_{pi}) + F_{si} + F_{ri} + \left(1 - \frac{\rho}{\rho_p} \right) m_p g_i \quad (4)$$

$$I_p \frac{d\omega_{pi}}{dt} = T_i \quad (5)$$

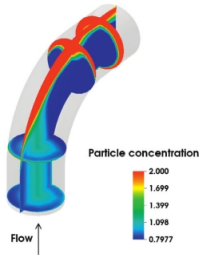




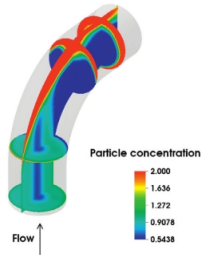
(a)



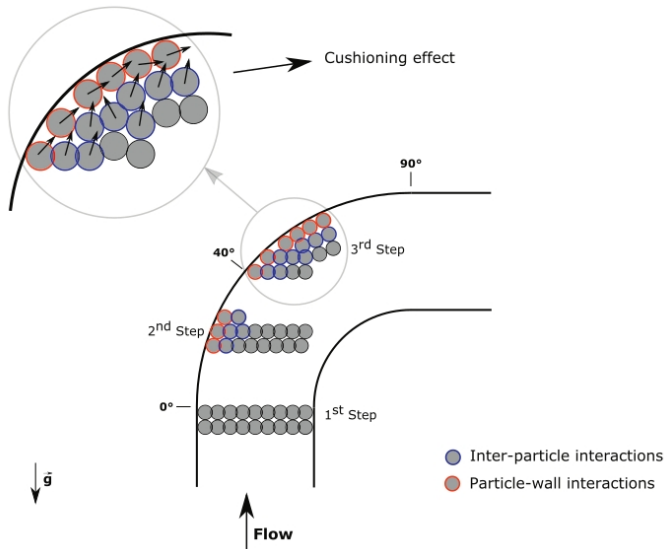
(b)

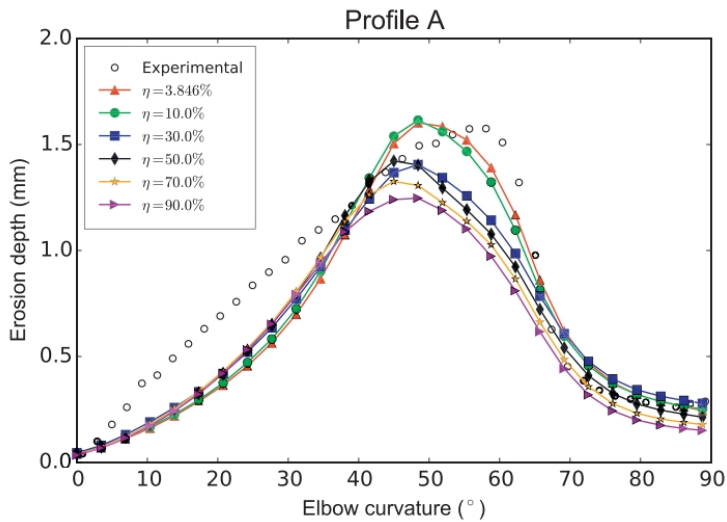


(a)



(b)







Article

CFD Modeling of Pressure Drop through an OCP Server for Data Center Applications

Aras Dogan ^{1,2}, Sibel Yilmaz ¹ , Mustafa Kuzay ^{1,2} , Cagatay Yilmaz ³  and Ender Demirel ^{1,2,*} 

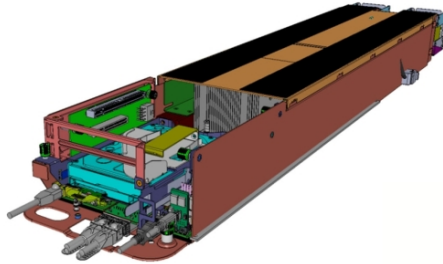
¹ Design and Simulation Technologies, Inc., 26480 Eskisehir, Turkey

² Department of Civil Engineering, Eskisehir Osmangazi University, 26480 Eskisehir, Turkey

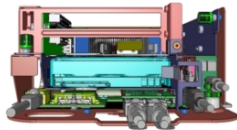
³ Lande Industrial Metal Products Inc. Co., Organized Industrial Zone, 20th Street, No: 14, 26110 Eskisehir, Turkey

* Correspondence: edemirel@ogu.edu.tr

(a)



(b)



(c)

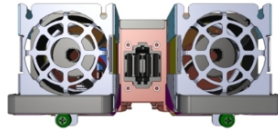


Figure 1. (a) Three-dimensional, (b) front and (c) back views of the Leopard V3.1 model of OCP server.

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\nu \frac{\partial u_i}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left(-\overline{u_i' u_j'} \right) + g_i + S_i \quad (2)$$

$$-\overline{u_i' u_j'} = \nu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}$$

$$\frac{\partial k}{\partial t} + u_i \frac{\partial k}{\partial x_i} = \frac{\partial}{\partial x_i} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial x_i} \right] + P - \beta k \omega \quad (4)$$

$$\frac{\partial \omega}{\partial t} + u_i \frac{\partial \omega}{\partial x_i} = \frac{\partial}{\partial x_i} \left[(\nu + \sigma_\omega \nu_t) \frac{\partial \omega}{\partial x_i} \right] + \alpha S^2 - \beta \omega^2 + 2(1 - F_1) \frac{\sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i} \quad (5)$$

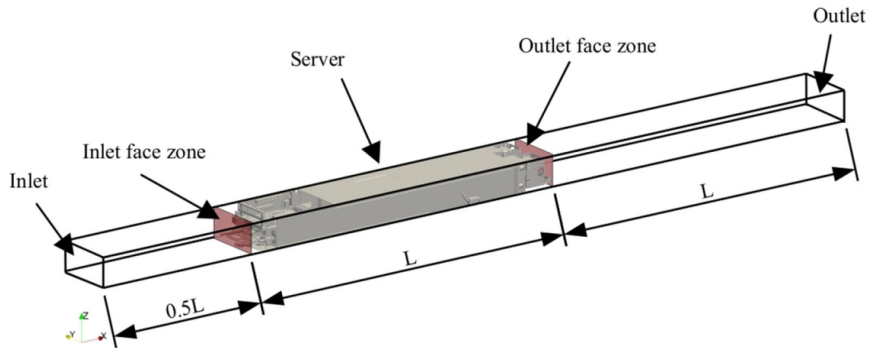
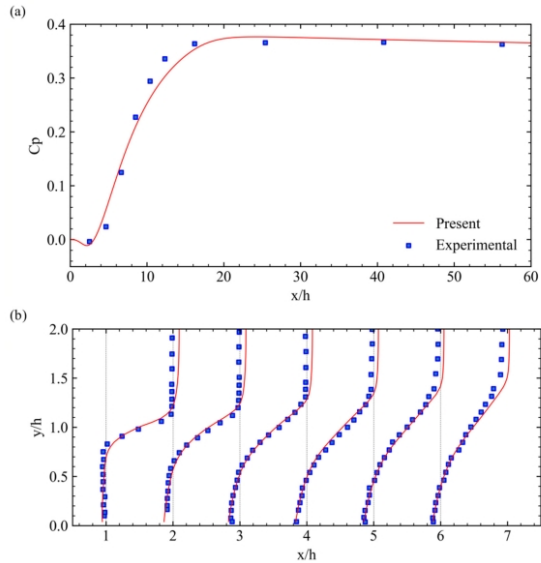


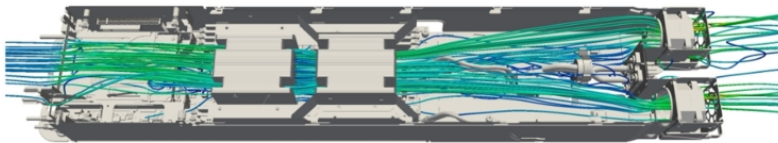
Figure 2. Computational domain and boundaries of the VWT.

Table 1. Boundary conditions of the VWT.

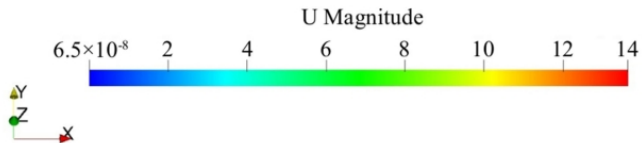
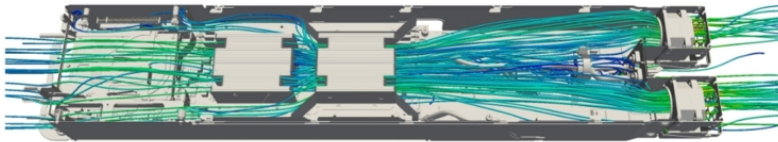
Inlet	Outlet	Slip	Walls
$u = U_0$	$\frac{\partial u}{\partial n} = 0$	$\frac{\partial u_i}{\partial n} = 0 = 0, i \perp n, u_n = 0$	$u = 0$
$\frac{\partial p}{\partial n} = 0$	$p = p_0$	$\frac{\partial p}{\partial n} = 0$	$\frac{\partial p}{\partial n} = 0$
$k = 1.5(I u)^2$	$\frac{\partial k}{\partial n} = 0$	$\frac{\partial k}{\partial n} = 0$	$\frac{\partial k}{\partial n} = 0$
$\omega = \frac{k^{0.5}}{C_\mu^{0.25} L}$	$\frac{\partial \omega}{\partial n} = 0$	$\frac{\partial \omega}{\partial n} = 0$	$\omega = \begin{cases} \omega_{vis} & y^+ \leq y^+_{laminar} \\ \omega_{log} & y^+ > y^+_{laminar} \end{cases}$ $\omega_{vis} = \frac{6\nu_\omega}{\beta_1 y^2}$ $\omega_{log} = \frac{\sqrt{k}}{C_\mu \kappa y}$
calculated	$\frac{\partial v_i}{\partial n} = 0$	$\frac{\partial v_i}{\partial n} = 0$	$v_t = \begin{cases} v \left(\frac{y^+ \kappa}{\ln E y^+} - 1 \right) & y^+ > y^+_{laminar} \\ 0 & y^+ \leq y^+_{laminar} \end{cases}$ $y^+_{laminar} = 3.8439$ $\kappa = 0.4$ $E = 9.8$

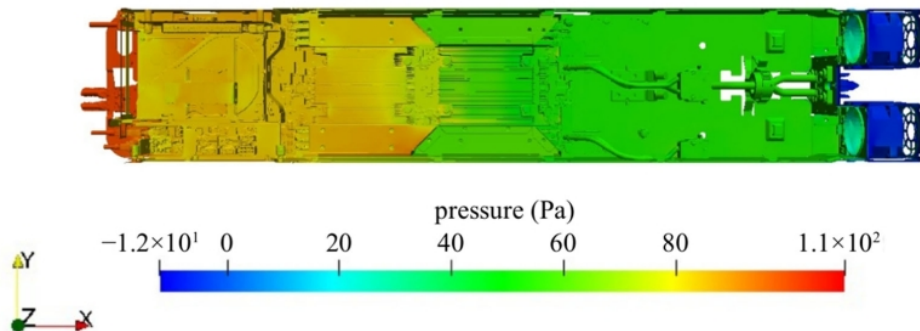


(a)



(b)





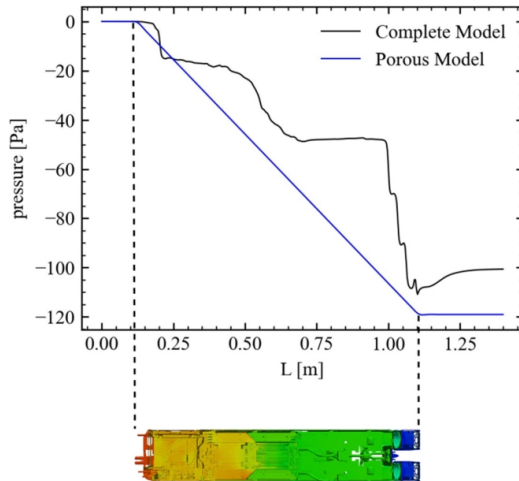


Figure 15. Comparison of the pressure distributions for the porous and complete models.

Numerical Simulation of Rotor-Stator Interaction Noise in Transonic Cascades

Roberto F. Bobenrieth Miserda,^{*} Braulio Gutierrez Pimenta,^{*} and Luiza Sampaio da Rocha[†]

University of Brasilia, 70910-900 Brasilia, Brazil

<https://doi.org/10.2514/1.B37627>

The first objective of this work is the verification of a moving-body immersed boundary method for the direct computation of noise generated by rotor-stator interactions in transonic 2-D cascades. This method is based on a discrete forcing approach where the boundary conditions are directly imposed at the surface of moving and static cascades immersed in a domain discretized by a regular Cartesian grid. The numerical method is verified using the mode generation and propagation theory applied to thin annular ducts for two-dimensional cascades. Two cases are simulated, the rotor alone and the rotor-stator interaction. For the first case, all computed modes are generated following the Tyler-Sofrin rule, but none of them are cut-on, as predicted by the duct propagation theory. For the second case, all modes are also generated in accordance with that rule, and the numerical results for the cut-on modes are also in excellent agreement with the duct propagation theory regarding mode number, signal, and angle of the wavefront. The second objective is to study the effect of decreasing rotor-stator separation on tone and broadband noise.



Fig. 9 Thin annular duct with 16 rotor blades (gray) and 13 stator vanes (black).

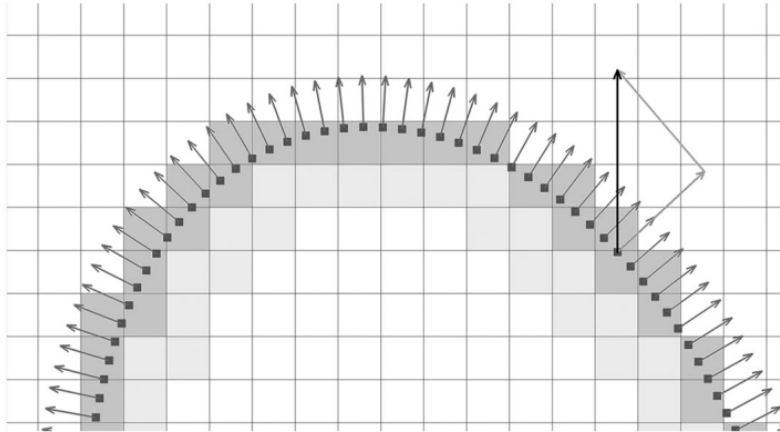
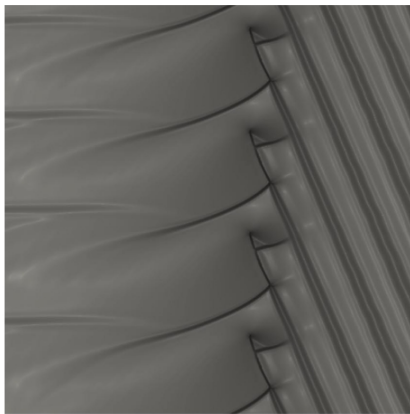
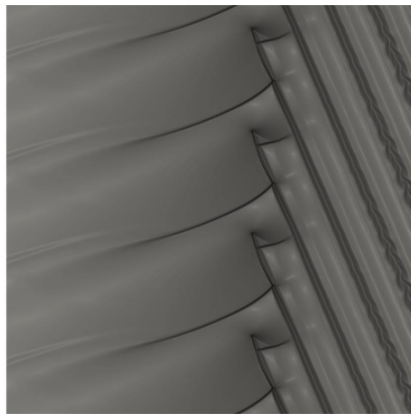


Fig. 1 Two-dimensional regular Cartesian grid showing the surface-grid points with normal unit vectors.



a) $\Delta x = \Delta y = 16h = 1/128$



b) $\Delta x = \Delta y = 8h = 1/256$

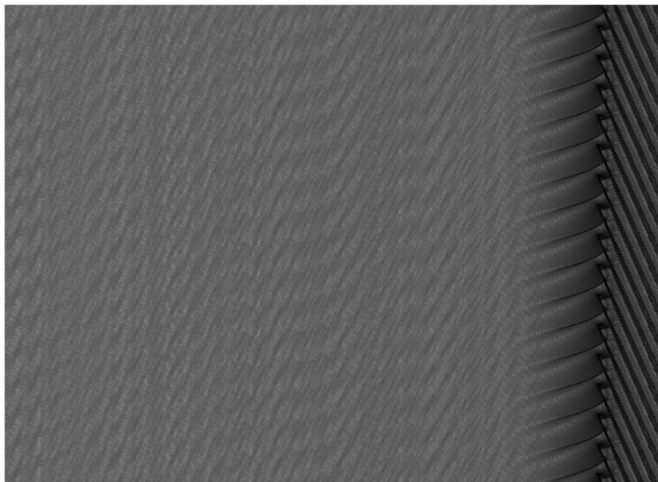
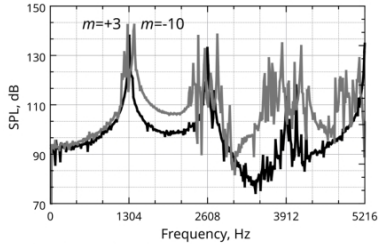
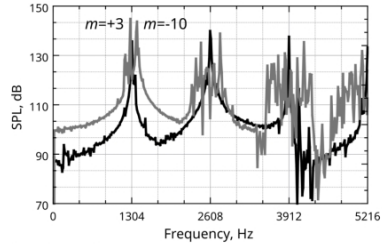


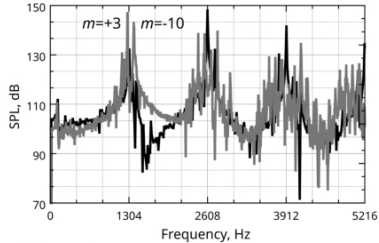
Fig. 15 Visualization of the instantaneous aeroacoustic field generated by the transonic rotor cascade.



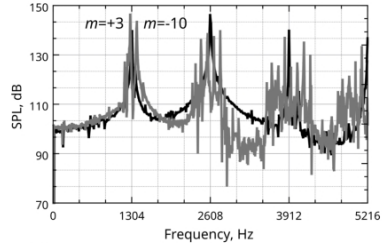
a) Maximum rotor-stator separation



b) 2/3 of maximum rotor-stator separation



c) 1/3 of maximum rotor-stator separation



d) Minimum rotor-stator separation

Fig. 24 Spectra of the moving (gray) and static (black) probes in the far-field position.



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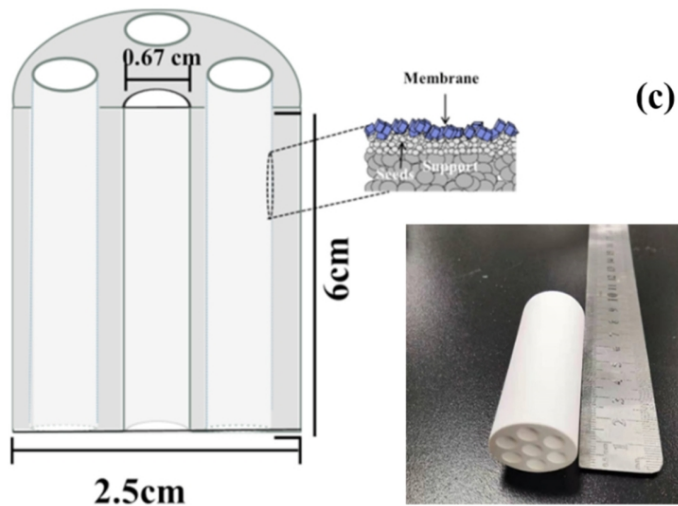
Review of zeolite membranes for natural gas processing and treatment: State of the art and future perspectives

Alysson M.A. Silva^{a, **}, Anna Carolina Britto de Faria^a, Camila de Lima Ribeiro^a,
Daniel D. Athayde^{a, b, *}, Edson Paulo da Silva^a, Giovana Magalhães dos Santos^a,
Luiz Fernando de Sousa Lima^a, Rodrigo Nunes de Souza^a, Saulo Lucas Pereira da Silva^a,
Nelcy D.S. Mohallem^c

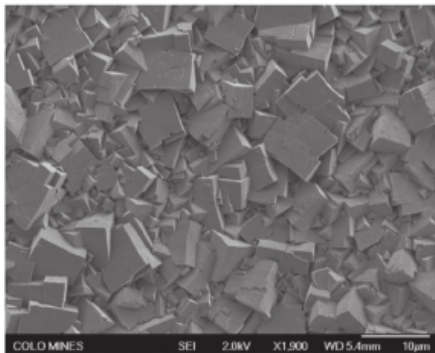
^a Mechanical Engineering Department, Faculty of Technology, Campus Universitário Darcy Ribeiro, University of Brasília, Brasília, 70910-900, DF, Brazil

^b Chemical Engineering Department, Universidade Federal de Minas Gerais - UFMG, Belo Horizonte, 31270-901, Brazil

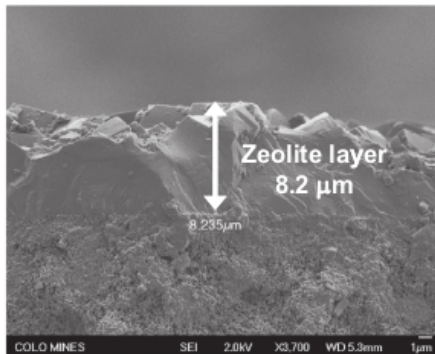
^c Chemistry department, Laboratório de Materiais Nanoestruturados, Universidade Federal de Minas Gerais – UFMG, Belo Horizonte, 37270-901, Brazil



a



b





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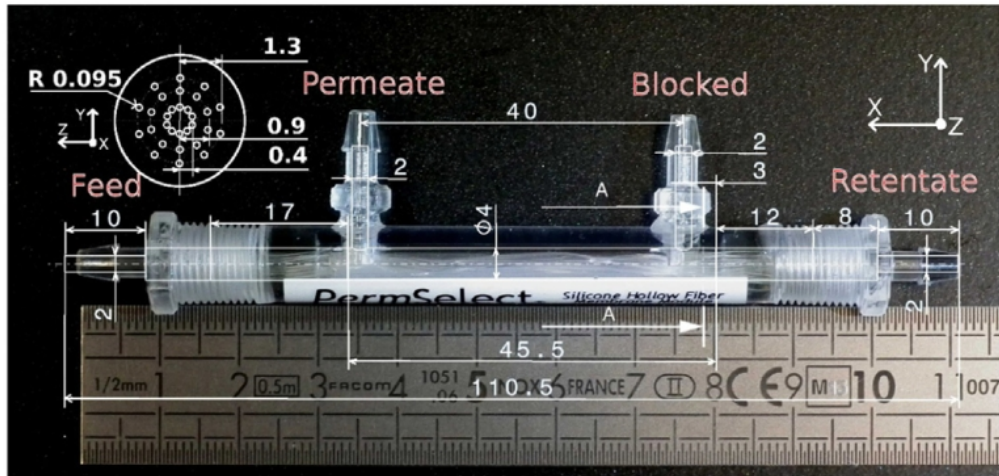
journal homepage: www.elsevier.com/locate/memsci

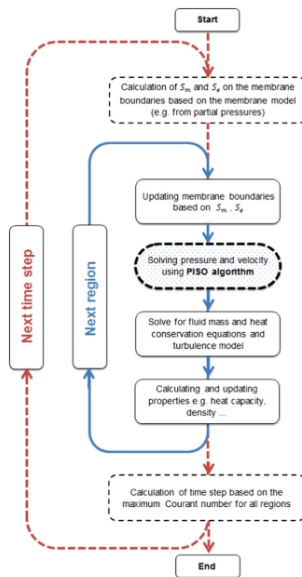
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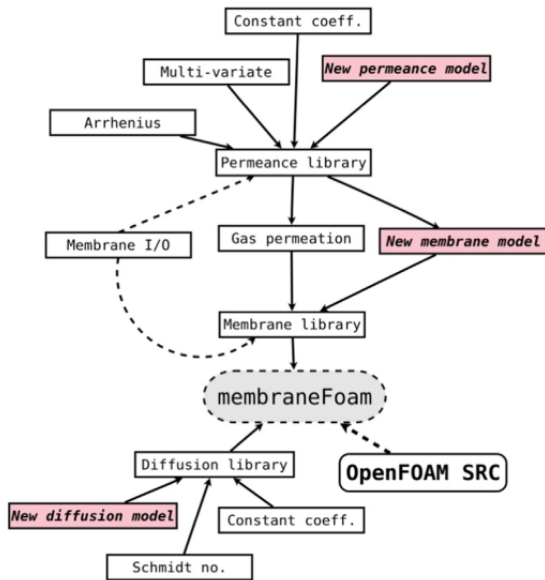


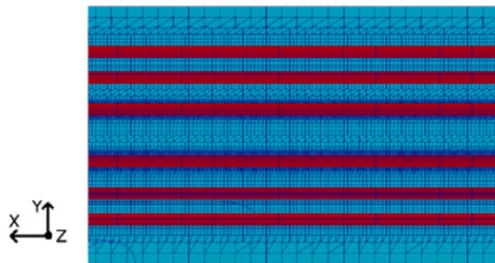
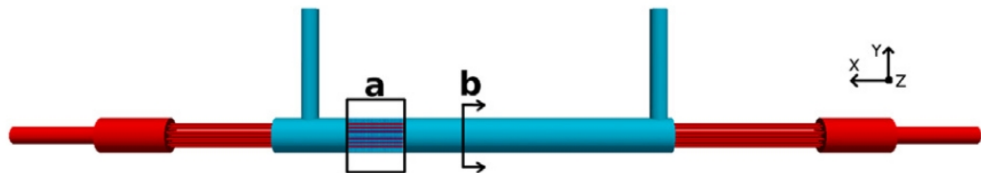
Bahram Haddadi*, Christian Jordan, Martin Miltner, Michael Harasek

Institute of Chemical, Environmental & Bioscience Engineering, TU Wien, Getreidemarkt 9/166, 1060 Vienna, Austria

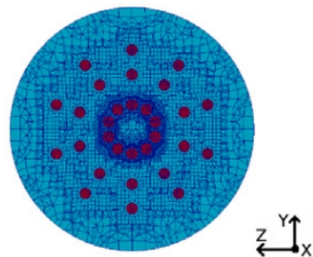




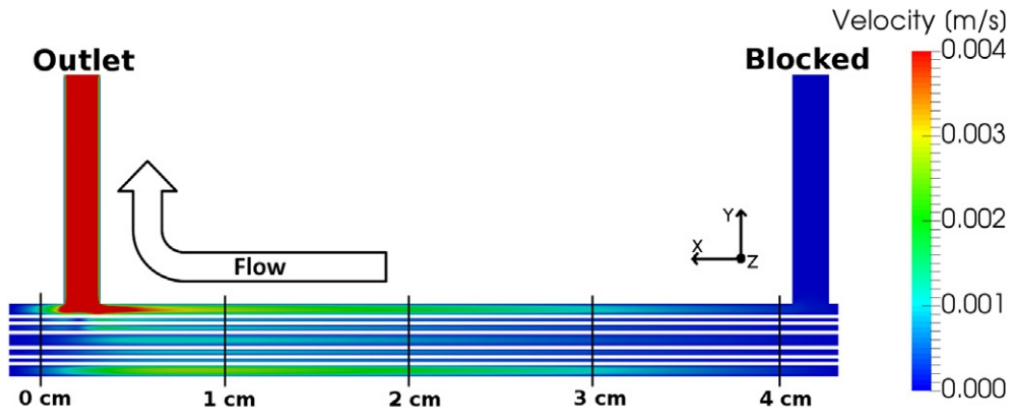


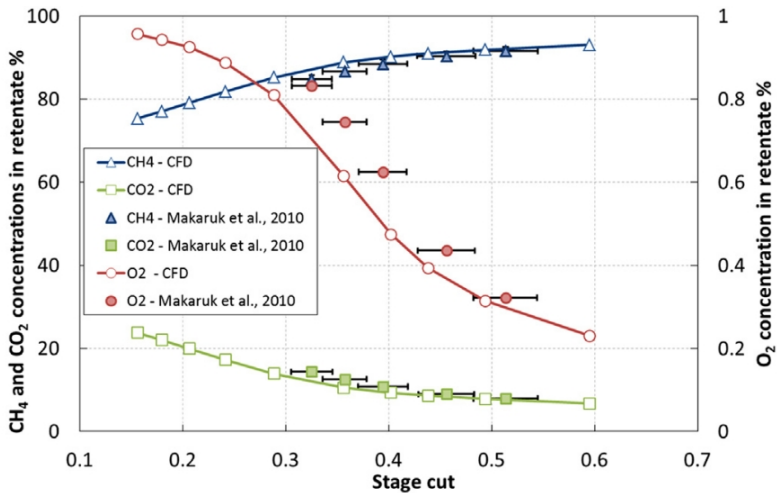


(a)



(b)







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Carotid artery hemodynamics before and after stenting: A patient specific CFD study



Michele Conti^{a,*}, Chris Long^c, Michele Marconi^d, Raffaella Berchiolli^d, Yuri Bazilevs^b,
Alessandro Reali^{a,e}

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^b Department of Structural Engineering, University of California, San Diego, 9500 Gilman Drive, La Jolla, CA, USA

^c Los Alamos National Laboratory, Los Alamos, NM, United States

^d IRCCS Cisanello Hospital, Pisa, Italy

^e Institute for Advanced Study, Technische Universität München, Lichtenbergstraße 2a, 85748 Garching, Germany

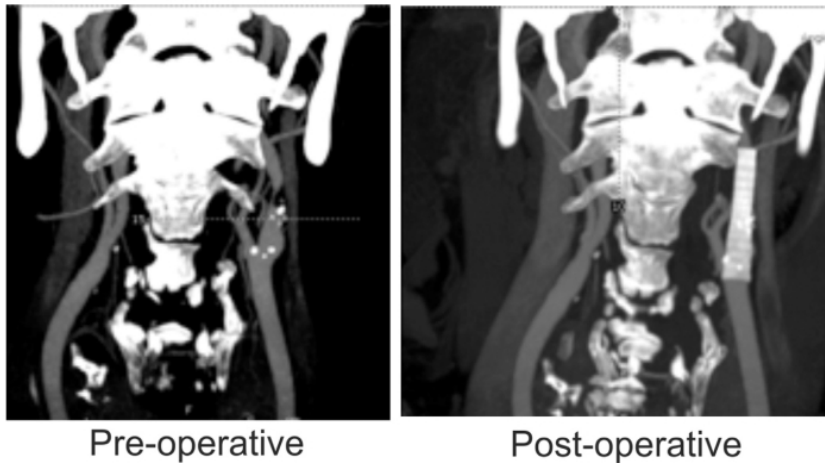
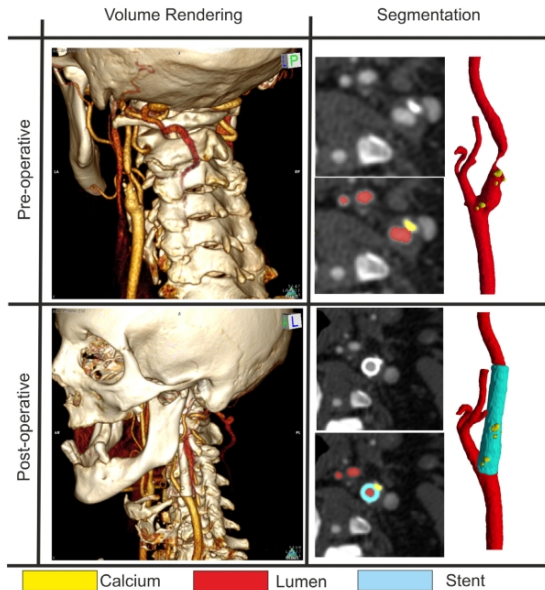
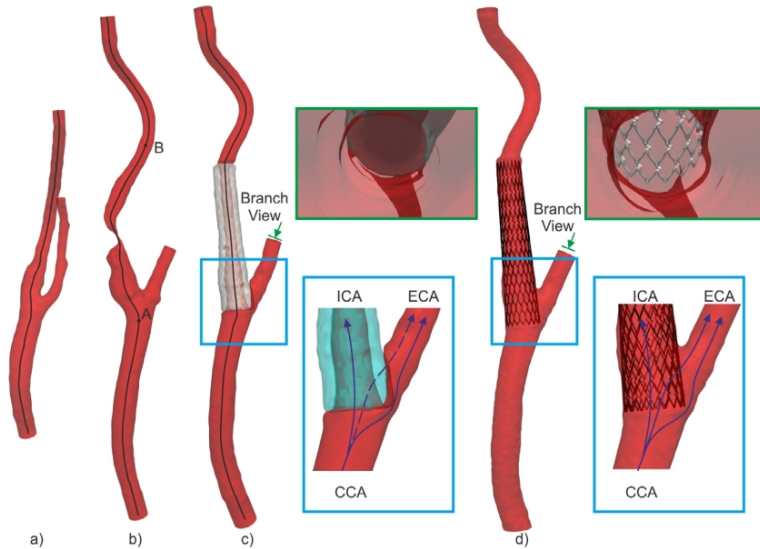
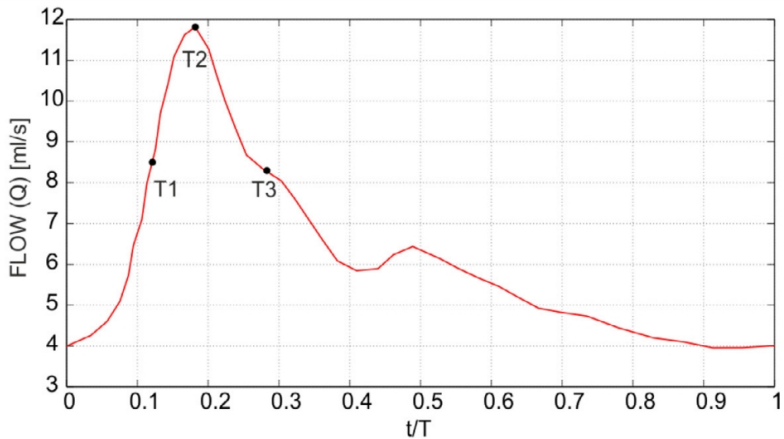


Fig. 1. Stenosis of the left carotid artery (pre-operative image) and deployed stent restoring vessel patency (post-operative image).







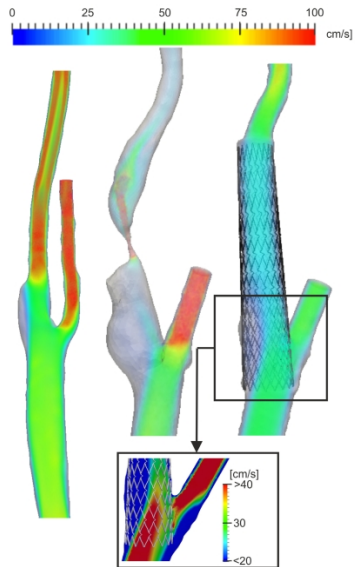


Fig. 11. Volume rendering of the magnitude of the computed velocity field for the healthy (left), pre-operative (middle), and post-operative cases at the systolic peak.

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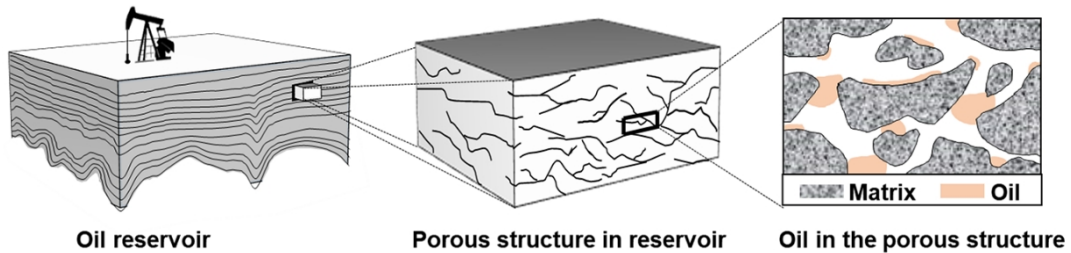
Pore-scale characteristics of multiphase flow in heterogeneous porous media using the lattice Boltzmann method

Received: 6 August 2018

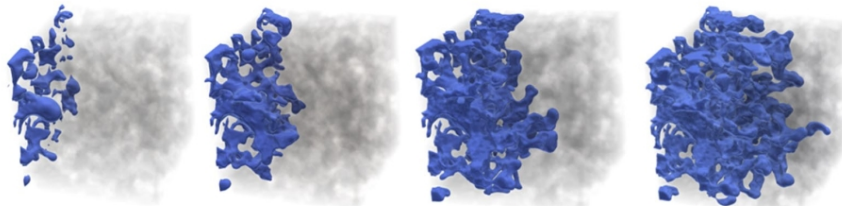
Accepted: 28 January 2019

Published online: 04 March 2019

Sahar Bakhshian¹, Seyyed A. Hosseini¹ & Nima Shokri² 



(a)



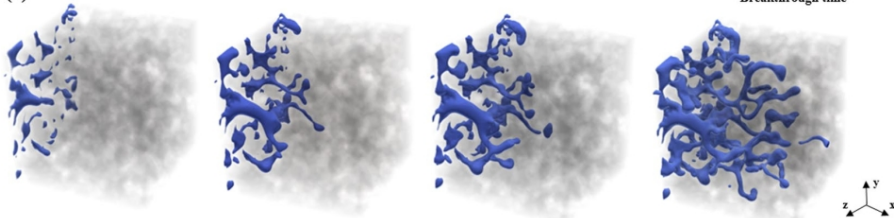
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200,000 Time-step

500,000 Time-step

1,128,000 Time-step
Breakthrough time

(b)

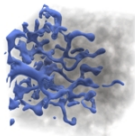


10,000 Time-step

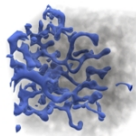
30,000 Time-step

40,000 Time-step

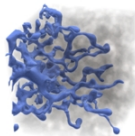
68,000 Time-step
Breakthrough time



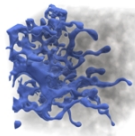
Case 1
Viscous fingering



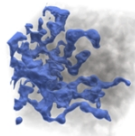
Case 2
Viscous fingering



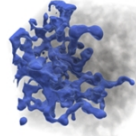
Case 3
Viscous fingering



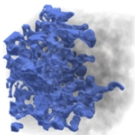
Case 4
Viscous fingering



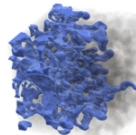
Case 5
Viscous fingering



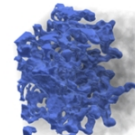
Case 6
Viscous fingering



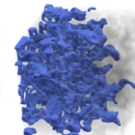
Case 7
Capillary fingering



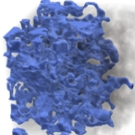
Case 8
Capillary fingering



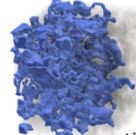
Case 9
Capillary fingering



Case 10
Stable displacement



Case 11
Stable displacement

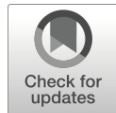


Case 12
Stable displacement



Computational Geosciences (2020) 24:1087–1100
<https://doi.org/10.1007/s10596-019-09927-3>

ORIGINAL PAPER



Production optimization using a modified net present value

Eugênio Libório Feitosa Fortaleza¹  · Emanuel Pereira Barroso Neto¹ · Marco Emílio Rodrigues Miranda¹

Received: 16 January 2019 / Accepted: 12 December 2019 / Published online: 5 February 2020
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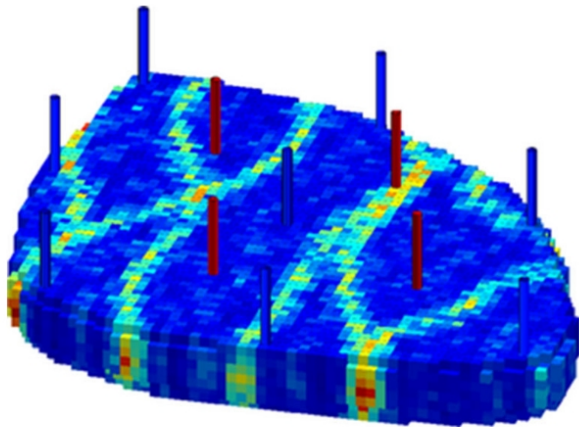


Fig. 1 A realization of Egg Model [19]

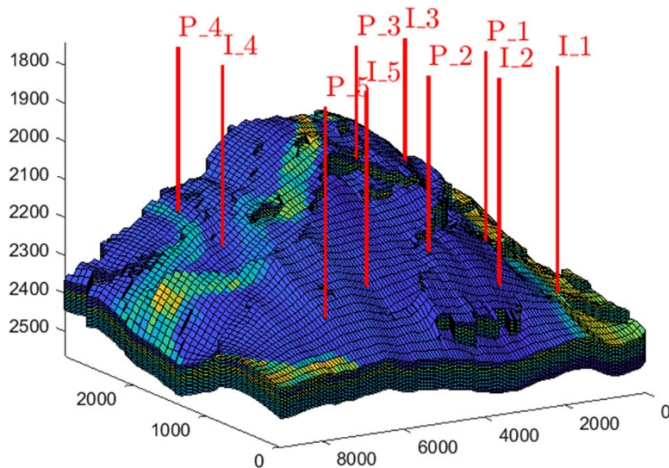
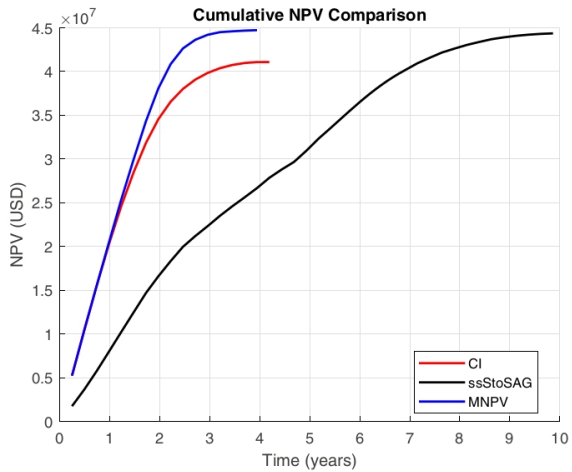


Fig.9 The SAIGUP model [20]

Table 3 Standard start point data for SAIGUP simulations

Description	Value
Initial production wells BHP (all wells)	300 bar
Initial injection wells rate (all wells)	5000 m ³ /day
Initial oil saturation	1.0
Initial water saturation	0.0
Simulation time	10 years
Time step	90 days
Variation parameter k_{δ}	1
Variation parameter κ	10 ⁻⁵
Variation parameter h	10 ⁻³



Theor. Comput. Fluid Dyn.
<https://doi.org/10.1007/s00162-020-00515-1>

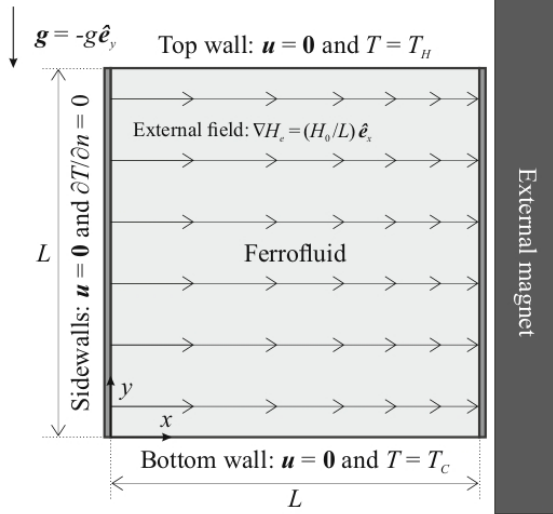


ORIGINAL ARTICLE

Lucas H. P. Cunha · Ivan R. Siqueira · Arthur A. R. Campos ·
Adriano P. Rosa · Taygoara F. Oliveira

A numerical study on heat transfer of a ferrofluid flow in a square cavity under simultaneous gravitational and magnetic convection

Received: 13 November 2019 / Accepted: 23 January 2020
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$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\rho \mathbf{u} \cdot \nabla \mathbf{u} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}, \quad (2)$$

$$\rho C_v \mathbf{u} \cdot \nabla T = k \nabla^2 T. \quad (3)$$

$$\mathbf{f} = \rho \beta (T_0 - T) \mathbf{g} + \mu_0 \chi_0 \beta_m (T_0 - T) \nabla \left(\frac{H^2}{2} \right). \quad (9)$$

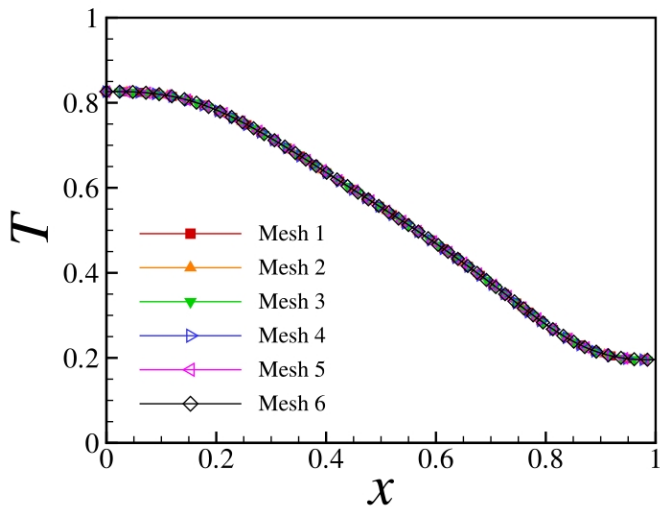


Table 2 Code validation: comparison with the results of de Vahl Davis [13] (Table V in his paper) for natural convection of air ($Pr = 0.71$) in a square cavity heated from the sides

	de Vahl Davis [13]				Present work			
	$Ra = 10^3$	$Ra = 10^4$	$Ra = 10^5$	$Ra = 10^6$	$Ra = 10^3$	$Ra = 10^4$	$Ra = 10^5$	$Ra = 10^6$
$\overline{Nu} (x = 0)$	1.118	2.243	4.519	8.800	1.118	2.245	4.522	8.825
$Nu_{\max} (x = 0)$	1.505	3.528	7.717	17.925	1.506	3.531	7.720	17.535
y-position	0.092	0.143	0.081	0.0378	0.090	0.142	0.080	0.0381
$Nu_{\min} (x = 0)$	0.692	0.586	0.729	0.989	0.691	0.585	0.728	0.980
y-position	1	1	1	1	1	1	1	1
$u_{\max} (x = 0.5)$	3.649	16.178	34.73	64.63	3.649	16.182	34.73	64.83
y-position	0.813	0.823	0.855	0.850	0.813	0.822	0.852	0.850
$v_{\max} (y = 0.5)$	3.697	19.617	68.59	219.36	3.697	19.622	68.60	220.54
x-position	0.178	0.119	0.066	0.0379	0.178	0.116	0.068	0.0381

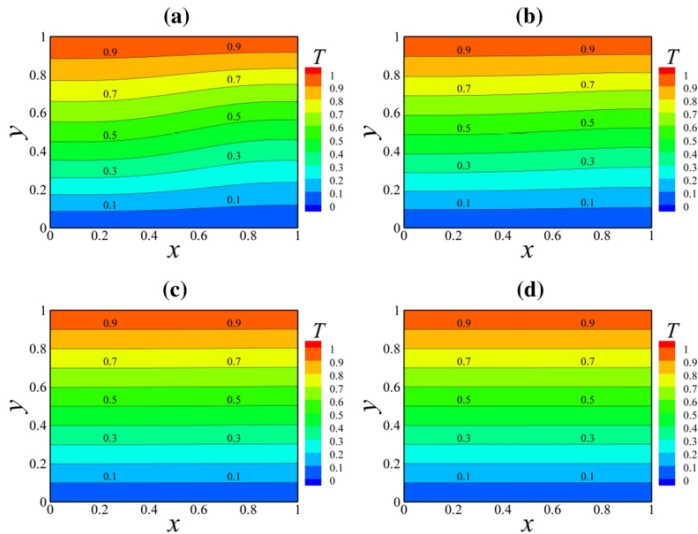


Fig. 4 Temperature field and isotherms at $Ra_m = 10^3$: **a** $Ra = 10^3$; **b** $Ra = 10^4$; **c** $Ra = 10^5$; **d** $Ra = 10^6$

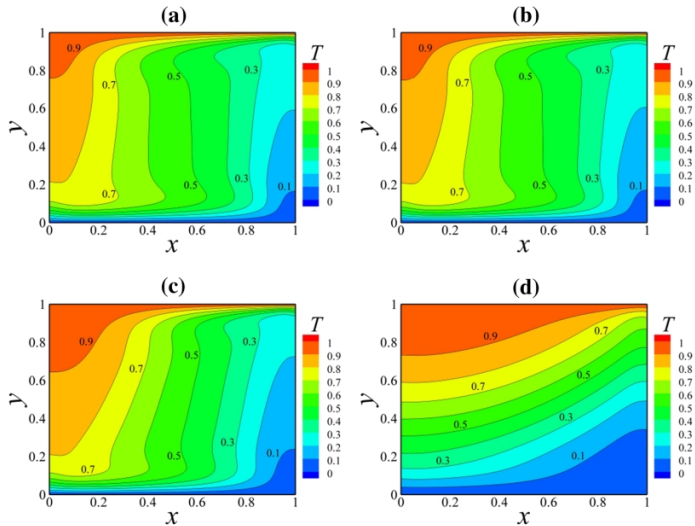


Fig. 5 Temperature field and isotherms at $Ra_m = 10^6$: **a** $Ra = 10^3$; **b** $Ra = 10^4$; **c** $Ra = 10^5$; **d** $Ra = 10^6$

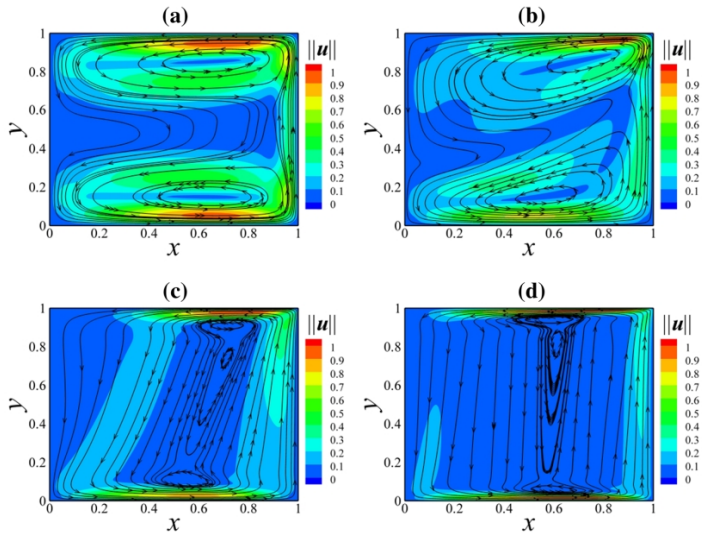
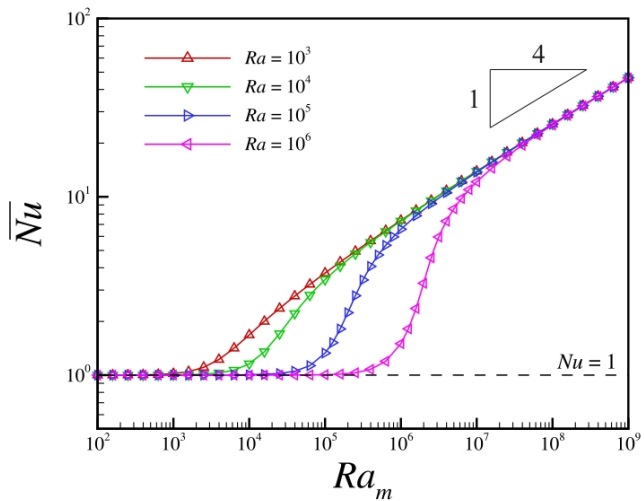
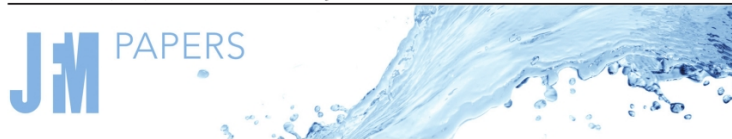


Fig. 7 Magnitude of the velocity field (normalized by its maximum value in the flow) and streamlines at $Ra = 10^6$: **a** $Ra_m = 10^5$ ($||\mathbf{u}||_{\max} = 3.58$); **b** $Ra_m = 10^6$ ($||\mathbf{u}||_{\max} = 44.68$); **c** $Ra_m = 10^7$ ($||\mathbf{u}||_{\max} = 551.68$); **d** $Ra_m = 10^8$ ($||\mathbf{u}||_{\max} = 1805.19$)





On the rheology and magnetization of dilute magnetic emulsions under small amplitude oscillatory shear

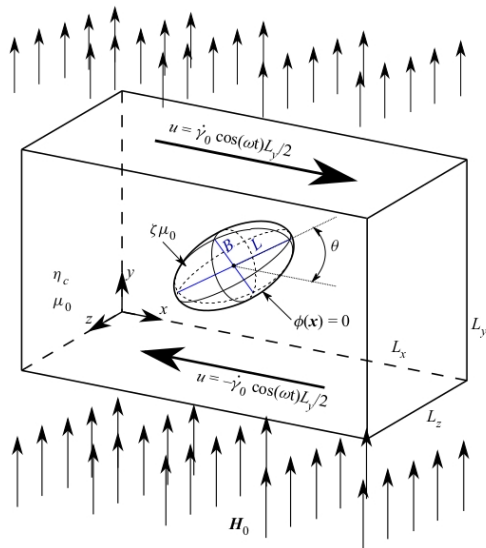
Rodrigo F. Abdo^{1,2,†}, Victor G. Abicalil^{1,†}, Lucas H.P. Cunha^{3,4,†} and Taygoara F. Oliveira^{1,†}

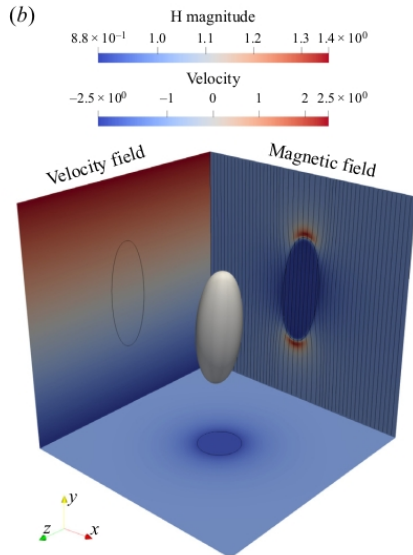
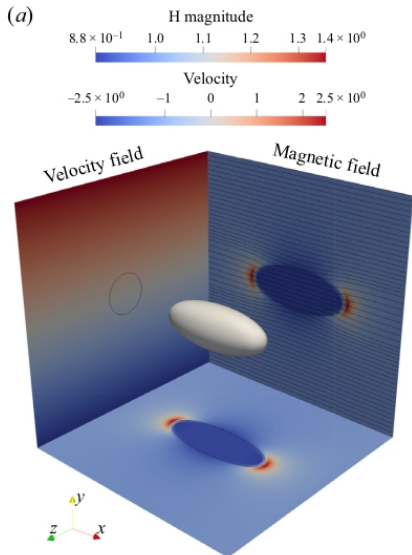
¹Laboratory of Energy and Environment, Department of Mechanical Engineering, University of Brasília, Brasília, DF 70910-900, Brazil

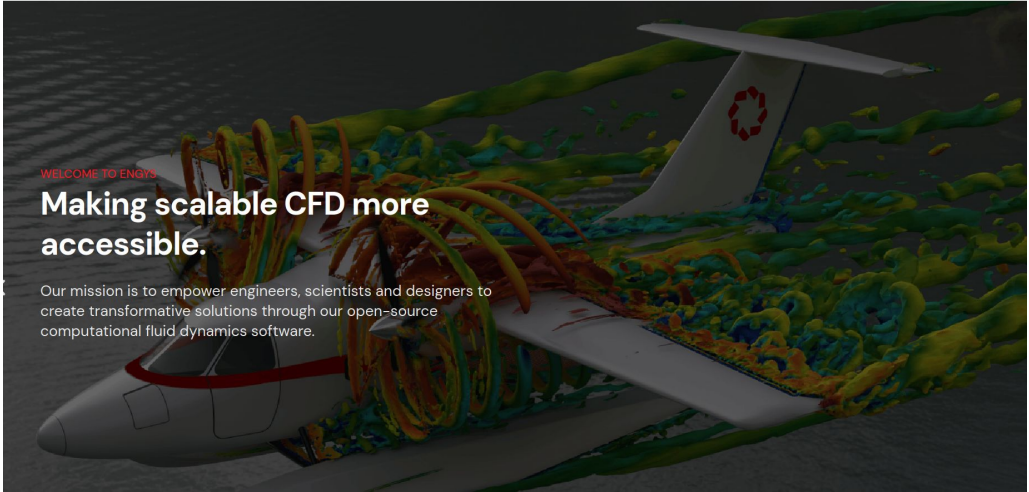
²Federal Institute of Brasília - Estrutural Campus, Brasília, DF 71250-000, Brazil

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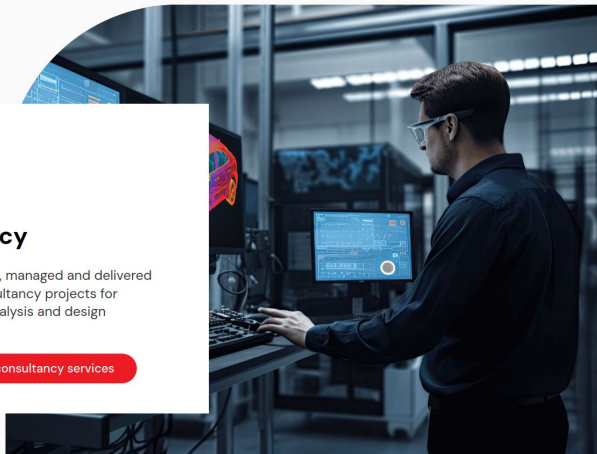
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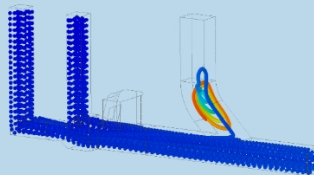
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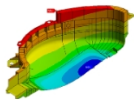
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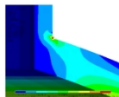
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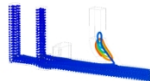
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- 3 OpenFOAM
- 4 Plano de Ensino
- 5 Material de Apoio
- 6 Trabalhos
- 7 Tarefa de Casa

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version 11

11th July 2023

<https://openfoam.org>



master

OpenFOAM-11 / applications / modules / incompressibleFluid / momentumPredictor.C

Code

Blame

59 lines (45 loc) · 1.86 KB

```
31
32 void Foam::solvers::incompressibleFluid::momentumPredictor()
33 {
34     volVectorField& U(U_);
35
36     tUEqn =
37     (
38         fvm::ddt(U) + fvm::div(phi, U)
39         + MRF.DDt(U)
40         + momentumTransport->divDevSigma(U)
41         ==
42         fvModels().source(U)
43     );
44     fvVectorMatrix& UEqn = tUEqn.ref();
45
46     UEqn.relax();
47
48     fvConstraints().constrain(UEqn);
49
50     if (pimple.momentumPredictor())
51     {
52         solve(UEqn == -fvc::grad(p));
53
54         fvConstraints().constrain(U);
55     }
56 }
```

Sumário

- 1 CFD
- 2 Softwares CFD
- 3 OpenFOAM
- 4 Plano de Ensino**
- 5 Material de Apoio
- 6 Trabalhos
- 7 Tarefa de Casa

Plano de Ensino.

Objetivos deste curso:

- familiarização com CFD e com o OpenFOAM;
- elaboração de um artigo científico.

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CFD Online

OpenFOAM wiki

Youtube: OpenFOAM Journal

Curso de OpenFOAM LabCC - UFSC

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Trabalho 1.

Cavidade cisalhante. Junção T. Escoamento em torno de cilindro.

Trabalho 2.

Convecção Natural. Cilindro com placa. Degrau. Modelos de turbulência.

Trabalho Final.

Artigo científico.

Data de entrega definida no Plano de Ensino.

Não será possível entregar o trabalho após a data definida.

Trabalho Final: último dia de aula.

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Tarefa de casa (para a próxima aula):

- entrar no site de cada um dos softwares CFD listados;
- fazer o cadastro no CFD Online;
- instalar OpenFOAM 11;
- instalar o Gmsh;
- propor um problema para ser o seu trabalho final.

Recomendação: instale o Ubuntu 24.04 LTS ou o Lubuntu 24.04 LTS.